Measures of Center & Spread

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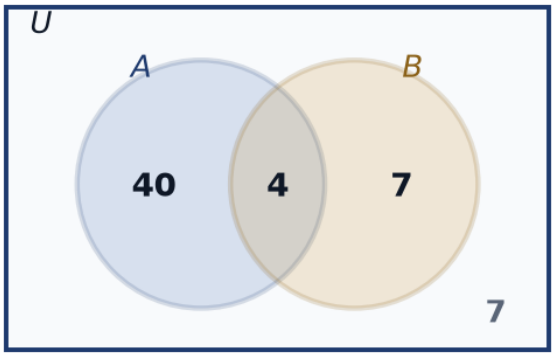
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Mean and median

Find the mean and median of the data set: $4, 7, 7, 9, 13$.

2 [Foundational] • Target: Mean and median

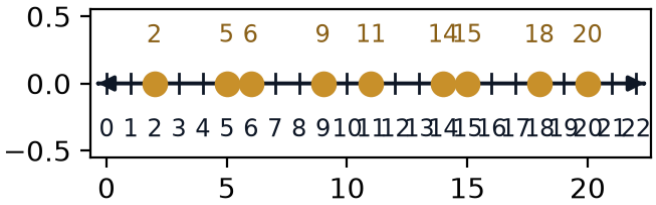
A sixth data point of 40 is added to the set $4, 7, 7, 9, 13$. Recompute the mean and median, and explain which measure changed more.



3 [Foundational] • Target: Range, IQR, and quartiles

Find the five-number summary (min, Q_1, median, Q_3, max) of the ordered data:

$2, 5, 6, 9, 11, 14, 15, 18, 20$.



4 [Foundational] • Target: Range, IQR, and quartiles

Using the five-number summary from the previous problem, find the IQR and determine whether 20 would be flagged as an outlier using the $1.5 \times$ IQR rule.

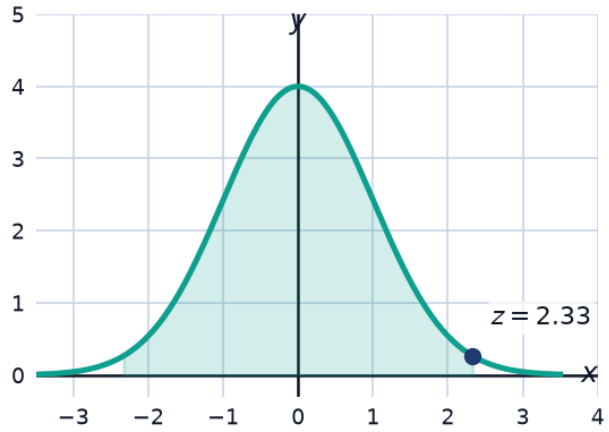
5 [Foundational] • Target: Standard deviation and variance

Find the standard deviation of the data set $2, 4, 6, 8$ (treat as a population).

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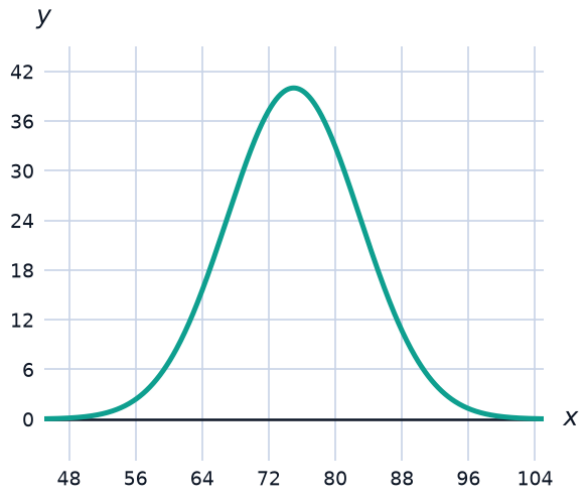
6 [Foundational] • Target: Standard deviation and variance

Explain why the standard deviation is always non-negative, and why it equals 0 only in one specific case.



7 [Foundational] • Target: Visualizing spread with a normal-shaped distribution

The graph shows a bell-shaped distribution of exam scores with mean 75 and standard deviation 8.



a Estimate the range of scores that fall within one standard deviation of the mean.

b Estimate the range of scores that fall within two standard deviations of the mean.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Comparing distributions using measures of spread

Class A has a mean test score of 80 with a standard deviation of 4. Class B has a mean of 80 with a standard deviation of 12. Compare the two classes' consistency, and explain what a small vs. large standard deviation indicates.

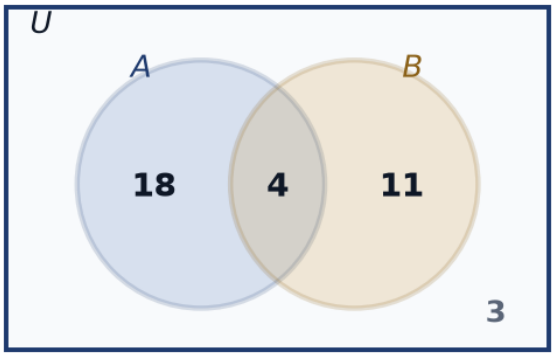
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9 [Intermediate] • Target: Resistant vs. non-resistant statistics

Explain why the median and IQR are described as "resistant" measures, while the mean and standard deviation are not, using the outlier example from earlier in this worksheet as a reference.

10 [Intermediate] • Target: Choosing appropriate summary statistics

A data set of home prices in a neighborhood is strongly skewed right due to a few very expensive mansions. Recommend which pair of summary statistics (mean/standard deviation or median/IQR) should be reported, and justify your choice.

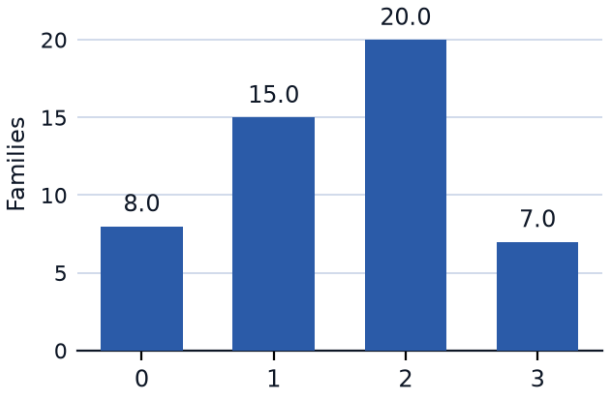


11 [Intermediate] • Target: Choosing appropriate summary statistics

A quality-control process measures the diameter of manufactured bolts, which are expected to be very consistent (low variability) and roughly symmetric. Recommend which pair of summary statistics is more appropriate here, and explain why.

12 [Intermediate] • Target: Computing summary statistics from a frequency table

A survey of families records number of children per household: 0 children (8 families), 1 child (15 families), 2 children (20 families), 3 children (7 families). Find the mean number of children per household.



13 [Intermediate] • Target: Computing summary statistics from a frequency table

A small business records weekly sales (in thousands): $10, 12, 12, 14, 17$. Find the mean, and then find the variance and standard deviation (treat as a population).

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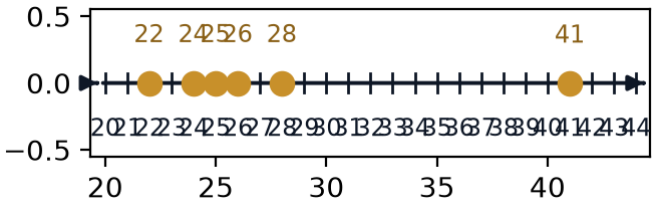
14 [Intermediate] • Target: Computing summary statistics from a frequency table

Two data sets both represent quiz scores out of 20 for two different classes of the same size, and both have the same mean of 15. Class A has a standard deviation of 1.2, while Class B has a standard deviation of 4.5. Without doing any further calculation, describe what a dotplot of each class's scores would likely look like, and explain how you can tell just from the two standard deviations.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

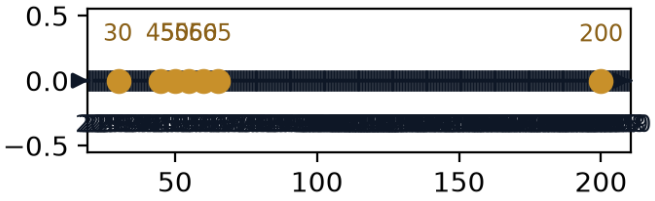
15 [Challenging] • Target: Computing summary statistics from a frequency table • Cross-Lesson: 3.01 Probability Rules & Conditional Probability

A coach records the vertical jump heights (in inches) of 7 athletes: $22, 24, 25, 26, 26, 28, 41$. Explain why the value 41 would likely be flagged as an outlier, and describe, in general terms, the two-step process for using the $1.5 \times$ IQR rule to formally check whether a value is an outlier.



16 [Challenging] • Target: Computing summary statistics from a frequency table • Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions

A teacher records the number of minutes 8 students spent on homework last night: $30, 45, 50, 55, 60, 60, 65, 200$. The value 200 comes from a student who mistakenly reported total weekly homework time instead of just last night. Explain, step by step, how this single incorrect data entry would affect the reported mean versus the reported median, and explain which measure a teacher should trust more when summarizing typical homework time for this data set, given the known data-entry error.



17 [Challenging] • Target: Mean and median • Cross-Lesson: 3.03 The Normal Distribution

Level 2 · MediumThe heights of adult males in a region follow a normal distribution with mean $\mu = 70$ inches and standard deviation $\sigma = 3$ inches. What is the standardized z-score of a male who is 76 inches tall, and what proportion of adult males are taller than 76 inches?

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18 [Challenging] • Target: Range, IQR, and quartiles • Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem

The commute times for employees at a large tech company have a strongly right-skewed distribution with mean $\mu = 38$ minutes and standard deviation $\sigma = 15$ minutes. A random sample of 49 employees is selected. (a) Describe the shape, mean, and standard error of the sampling distribution of the sample mean $\bar{x}$. (b) Calculate $P(\bar{x} > 42 \text{ minutes})$.

19 [Challenging] • Target: Standard deviation and variance • Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing

Level 3 · AdvancedAn assembly line fills cereal boxes with a mean weight $\mu = 450$ grams and standard deviation $\sigma = 18$ grams. A random sample of $n = 36$ boxes is selected. What is the mean and standard error of the sample mean weight $\bar{x}$, and what is the probability that $\bar{x}$ exceeds 455 grams?

20 [Challenging] • Target: Visualizing spread with a normal-shaped distribution • Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference

A set of daily high temperatures in Celsius has mean $\bar{x} = 25.0^\circ\text{C}$ and standard deviation $s = 4.0^\circ\text{C}$. Convert the temperatures to Fahrenheit using $F = 1.8C + 32$. Find the new mean and standard deviation in Fahrenheit.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Mean and median

Find the mean of the data set: $17, 28, 12, 6, 13$.

a. $\frac{81}{5}$

b. $\frac{71}{5}$

c. 19

d. $\frac{76}{5}$

22 [Foundational] • Target: Range, IQR, and quartiles

Find the median of the data set: $16, 15, 5, 25, 29, 12$.

a. $\frac{31}{2}$

b. 17

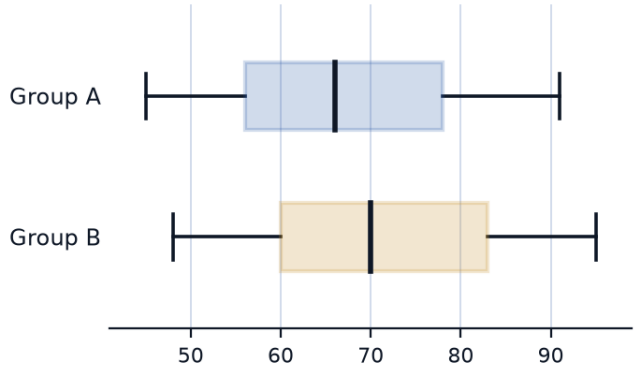
c. $\frac{33}{2}$

d. $\frac{37}{2}$

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23 [Foundational] • Target: Standard deviation and variance

Find the IQR of the ordered data set: $4, 7, 20, 29, 43, 50, 51, 54$.



a. 36

b. $\frac{101}{2}$

c. 38

d. 37

24 [Foundational] • Target: Visualizing spread with a normal-shaped distribution

A data set has $Q_1 = 22$ and $Q_3 = 43$. Using the $1.5 \times$ IQR rule, find the upper fence used to detect outliers.

a. $\frac{147}{2}$

b. 64

c. $\frac{149}{2}$

d. $\frac{151}{2}$

25 [Foundational] • Target: Comparing distributions using measures of spread

Find the standard deviation of the data set $14, 10, 13, 23$ (treat as a population), to two decimal places.

a. 5.15

b. 4.85

c. 4.55

d. 5.45

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26 [Foundational] • Target: Resistant vs. non-resistant statistics

A machine dispenses a liquid into containers. The amount of liquid dispensed, in milliliters, follows a continuous uniform distribution over the interval $[95, 105]$. A quality control inspector randomly selects 5 containers and measures the amount of liquid in each. Let X_i be the amount of liquid in the ith container, for $i = 1, 2, 3, 4, 5$. The company has implemented a new calibration process that, on average, increases the mean amount of liquid dispensed by δ milliliters, but also increases the variance by σ^2. The inspector is specifically looking for evidence that the calibration process has *decreased* the variability of the dispensed liquid while maintaining a satisfactory average fill. If the current calibration process yields a mean fill of $\mu_0 = 100$ ml and a standard deviation of $\sigma_0 = 2$ ml, and the new process is hypothesized to have a mean fill of $\mu_1 = 100 + \delta$ ml and a standard deviation of $\sigma_1 = 2 - \sigma_{dev}$ ml, where $\delta > 0$ and $\sigma_{dev} > 0$ are unknown positive constants, what is the minimum value of σ_{dev} such that the probability of falsely concluding that the new process has *increased* variability (i.e., Type I error) is at most 0.01, assuming the mean fill remains at 100 ml (i.e., $\delta = 0$)?

a. The probability cannot be determined without knowing the specific value of δ.

b. Approximately 1.01

c. Approximately 1.37

d. Approximately 2.00

27 [Foundational] • Target: Choosing appropriate summary statistics

The scores on a standardized test have a symmetric, bell-shaped distribution with mean 500 and standard deviation 100. Approximately what percentage of test takers scored between 400 and 600?

a. 50%

b. 68%

c. 95%

d. 99.7%

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28 [Foundational] • Target: Computing summary statistics from a frequency table

A pharmaceutical company is conducting a clinical trial to test the efficacy of a new drug designed to lower blood pressure. They recruit 200 participants and randomly assign them to either the treatment group (receiving the new drug) or the control group (receiving a placebo). After 12 weeks, they measure the reduction in systolic blood pressure for each participant. The company wants to perform a hypothesis test to determine if there is a significant difference in blood pressure reduction between the two groups. The trial is designed with 20 participants in the treatment group and 180 participants in the control group. If the standard deviation of blood pressure reduction in the treatment group is $\sigma_1 = 15$ mmHg and in the control group is $\sigma_2 = 12$ mmHg, and they are planning to use a two-sample t-test, what is the calculated pooled standard deviation s_p if equal variances are assumed, given the following summary statistics for blood pressure reduction (in mmHg): $\bar{x}_1 = 10$ mmHg for the treatment group and $\bar{x}_2 = 7$ mmHg for the control group?

a. $\sqrt{\frac{15^2 + 12^2}{2}}$

b. $\sqrt{\frac{20(15^2) + 180(12^2)}{200}}$

c. $\frac{20 \times 15 + 180 \times 12}{200}$

d. $\sqrt{\frac{19(15^2) + 179(12^2)}{198}}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 **[Intermediate]** • **Target: Mean and median**

A biostatistician is developing a new statistical test for comparing the proportion of successful treatments across three independent groups. The test involves calculating a test statistic that follows a chi-square distribution under the null hypothesis. The null hypothesis involves comparing the proportions of a binary outcome (e.g., recovery or no recovery) across these three groups. The data is presented in a contingency table. If the contingency table has 3 rows representing the treatment groups and 5 columns representing the outcome categories (e.g., various stages of recovery), what is the expected value of the test statistic under the null hypothesis, assuming the expected cell counts are all greater than or equal to 5, and given that the overall proportion of success across all groups is $p = 0.7$? Assume for the purpose of this problem that the true proportions for each group are known and lead to a specific deviation from the null, such that the sum of the squared differences between observed and expected counts, divided by the expected counts, is calculated. Consider a hypothetical scenario where the observed proportions for the three groups lead to expected cell counts that perfectly align with the grand proportion $p = 0.7$ for the 'success' column and $1 - p = 0.3$ for the 'no success' column. If the total sample size across all groups is $N = 150$, and the degrees of freedom for the chi-square test are df, what is the value of $\frac{1}{df}\sum\frac{(O_{ij} - E_{ij})^2}{E_{ij}}$ where O_{ij} are the observed counts and E_{ij} are the expected counts under the null hypothesis that group proportions are equal?

a. 0.7

b. 1.0

c. 1.5

d. 2.1

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30 [Intermediate] • Target: Range, IQR, and quartiles

A marketing research firm is designing an experiment to test the effectiveness of three different advertising slogans (Slogan A, Slogan B, Slogan C) on product purchase intent. They recruit 240 participants and randomly assign them to one of the three slogan groups. After exposure to their assigned slogan, participants rate their purchase intent on a scale of 1 to 10. The experiment is designed as a completely randomized design. The data collected will be analyzed using ANOVA to compare the mean purchase intent scores across the three groups. In order to determine the minimum number of participants needed in each group for a power analysis, the researchers need to estimate the effect size. A pilot study suggests that the standard deviation of purchase intent scores within each group is approximately $\sigma = 1.8$. They aim for a minimum of $n = 80$ participants per group to achieve the desired statistical power. If the total number of participants available for this experiment were $N = 240$ and they decided to allocate the participants into $k = 3$ treatment groups, with an equal number of participants in each group, what would be the number of participants per group, n? Additionally, if the experimenters decided to use a blocked design instead, where participants are first blocked by age (Young, Middle-aged, Senior) and then randomly assigned slogans within each block, how many blocks would be needed?

a. 80 participants per group; 3 blocks

b. 60 participants per group; 2 blocks

c. 80 participants per group; 2 blocks

d. 60 participants per group; 3 blocks

31 [Intermediate] • Target: Standard deviation and variance

A population has $\mu = 141$ and $\sigma = 30$. For a random sample of size $n = 36$, find $P(\bar{x} < 143)$ using the sampling distribution of the sample mean.

a. 0.7554

b. 0.9554

c. 0.3554

d. 0.6554

32 [Intermediate] • Target: Visualizing spread with a normal-shaped distribution

A population proportion is $p = 0.38$. For samples of size $n = 300$, find the standard deviation of the sampling distribution of $\hat{p}$.

a. 0.058

b. 0.028

c. 0.018

d. 0.068

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33 [Intermediate] • Target: Comparing distributions using measures of spread

A population has $\sigma = 30$. For random samples of size $n = 9$, find the standard deviation of the sampling distribution of $\bar{x}$ (the standard error).

a. 13

b. 6

c. 12

d. 10

34 [Intermediate] • Target: Resistant vs. non-resistant statistics

A distribution has mean $\mu = 71$ and standard deviation $\sigma = 8$. What is the z-score for $x = 55$?

a. -1.5

b. -2

c. -0.5

d. -2.5

35 [Intermediate] • Target: Choosing appropriate summary statistics

Which parameter describes spread in a Normal distribution?

a. Mean

b. Median

c. Standard deviation

d. Variance only

36 [Intermediate] • Target: Computing summary statistics from a frequency table

The box plot below shows the distribution of scores for a class on a recent statistics quiz. The five-number summary is given by minimum, first quartile (Q1), median, third quartile (Q3), and maximum. If the interquartile range (IQR) is 25, and the median score is 70, find the value of Q1.

a. 45

b. 60

c. 75

d. 95

37 [Intermediate] • Target: Mean and median

A data set of 20 distinct positive integers has a mean of 50 and a median of 45. If the largest number in the data set is increased by 100, which of the following statements must be true? I. The mean increases. II. The median increases. III. The standard deviation increases.

a. I only

b. I and II only

c. I and III only

d. I, II, and III

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Range, IQR, and quartiles • Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem

A pharmaceutical company is testing a new drug to reduce blood pressure. They conduct a clinical trial with $n = 100$ participants. The sample mean reduction in systolic blood pressure is $\bar{x} = 15$ mmHg with a sample standard deviation of $s = 8$ mmHg. The company plans to construct a 95% confidence interval for the true mean reduction in systolic blood pressure. If the margin of error for this confidence interval is $M = 2$ mmHg, what would be the required sample size, n', to achieve a margin of error of $M' = 1$ mmHg with the same confidence level and an estimated population standard deviation equal to the sample standard deviation?

a. 200

b. 300

c. 400

d. 1600

39 [Challenging] • Target: Standard deviation and variance • Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing

A software company develops a new encryption algorithm. The algorithm's security depends on a sequence of random integers generated by a custom pseudo-random number generator (PRNG). The PRNG is designed such that the probability of generating an integer in a specific range $[a, b]$ is proportional to the length of the range, $b - a$, for $0 \le a \le b \le 1$. Let X be the first integer generated and Y be the second integer generated by this PRNG, independently. The problem is to find the probability that the sum of the first two generated integers, $S = X + Y$, is less than 1. Assume the PRNG generates numbers from a continuous uniform distribution on $[0, 1]$.

a. 1/4

b. 1/3

c. 1/2

d. 2/3

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40 [Challenging] • Target: Visualizing spread with a normal-shaped distribution • Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference

A quality control engineer is monitoring the number of defects produced by two different machines, Machine A and Machine B. Let X be the number of defects produced by Machine A in a given hour, and Y be the number of defects produced by Machine B in the same hour. The joint probability distribution of X and Y is given in the table below: | $P(X = x, Y = y)$ | $y = 0$ | $y = 1$ | $y = 2$ | | :----------: | :-----: | :-----: | :-----: | | $x = 0$ | 0.10 | 0.15 | 0.05 | | $x = 1$ | 0.20 | 0.30 | 0.10 | | $x = 2$ | 0.05 | 0.05 | 0.00 | What is the expected value of the total number of defects produced by both machines in a given hour, $E(X + Y)$?

a. 1.15

b. 1.30

c. 1.45

d. 1.60

41 [Challenging] • Target: Comparing distributions using measures of spread • Cross-Lesson: 1.02 Z-Scores & Linear Transformations

A marketing research firm is analyzing the relationship between the preferred social media platform and age group of consumers. They collected data from a sample and summarized it in the two-way table below. The marginal distributions are also provided. The research team is interested in understanding how the distribution of preferred platform varies across age groups and is considering a chi-square test for independence. If the null hypothesis of independence is rejected, the observed counts will be significantly different from what would be expected under independence. Assume that a chi-square test for independence is to be performed on this data. Which of the following represents a potential challenge or insight derived from the structure of the data and the test's assumptions?

a. The number of degrees of freedom for the chi-square test can be determined solely from the total number of observations, irrespective of the table dimensions.

b. If the expected count in any cell is less than 5, a normal approximation to the binomial distribution for each cell count would be appropriate to proceed with the chi-square test.

c. The calculation of expected counts under the null hypothesis of independence relies on the marginal proportions of the observed data.

d. A high chi-square statistic indicates that the observed counts are very close to the expected counts, suggesting a strong association between age group and preferred social media platform.

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42 [Challenging] • Target: Resistant vs. non-resistant statistics • Cross-Lesson: 1.03 Describing Distributions & Linear Regression

A researcher is investigating the relationship between two categorical variables, 'Exercise Frequency' and 'Dietary Preference', using a chi-square test of independence. The contingency table summarizing the observed frequencies has 3 rows and 5 columns. What is the number of degrees of freedom df for this chi-square test?

a. 15

b. 12

c. 8

d. 6

43 [Challenging] • Target: Choosing appropriate summary statistics • Cross-Lesson: 2.01 Sampling Methods & Bias

Standard deviation of Binomial(n,p) is:

a. np

b. $np(1-p)$

c. $\sqrt{np(1-p)}$

d. $\sqrt{np}$

44 [Challenging] • Target: Computing summary statistics from a frequency table • Cross-Lesson: 2.02 Experimental Design & Scope of Inference

A variable follows a Normal distribution with $\mu = 78$ and $\sigma = 6$. Using the empirical rule, approximately what percent of values lie between 72 and 84?

a. 50%

b. 99.7%

c. 68%

d. 95%

45 [Challenging] • Target: Mean and median • Cross-Lesson: 3.01 Probability Rules & Conditional Probability

Two boxplots are compared: Group A has median 30 and IQR 17; Group B has median 43 and IQR 20. Which statement correctly compares their spread?

a. Group B has greater variability (larger IQR) than Group A

b. The group with the higher median always has less spread

c. Group A and Group B have the same median

d. Both groups are perfectly symmetric

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